

Haris Kamal Rana

+93-309-8999565 | harriskamal23@gmail.com | [linkedin.com/in/hariskamalrana](https://www.linkedin.com/in/hariskamalrana) | github.com/haristrana-dev

EDUCATION

The University of Lahore

Bachelor of Science in Computer Science (BSCS), CGPA: 3.66/4.00

Lahore, Pakistan

Sep. 2023 – Expected Jun. 2027

EXPERIENCE

Project Lead & Software Engineer – DriveVitals | [GitHub](#)

Jul. 2026 – Present

Self-Initiated Engineering Project

Remote

- Architected and led the development of **DriveVitals**, an AI-powered Fleet Intelligence and Digital Twin platform for simulating commercial fleet operations, vehicle telemetry, driver behaviour, and predictive fleet analytics.
- Designed the complete backend architecture, defining the Digital Twin execution pipeline from simulation runtime, fleet management, driver decision-making, vehicle control, physics simulation, telemetry generation, analytics, and dashboard integration.
- Implemented the core Digital Twin framework using Python and FastAPI, developing the runtime engine, manager architecture, domain entities, decision engine, vehicle controller, physics engine, and Version 1 backend APIs.
- Designed a modular enterprise-grade software architecture following SOLID principles, dependency injection, and scalable component-based design to support future OBD-II integration and machine learning models.
- Led a team of three developers by planning development milestones, assigning technical tasks, establishing Git workflows using feature branches and pull requests, reviewing code, and managing controlled merges into the development branch.
- Designed the PostgreSQL data model and backend architecture for fleet management, historical telemetry, trip tracking, driver performance, vehicle health, and predictive maintenance capabilities.

PROJECTS

Smart Door Security System | [GitHub](#) | *Python, OpenCV, Computer Vision*

2025

- Engineered a real-time facial recognition security system using Python and OpenCV for automated identity verification.
- Developed a complete computer vision pipeline including face detection, feature encoding, recognition matching, and access decision logic.
- Implemented head pose-based liveness detection with randomized directional prompts to prevent spoofing attacks.
- Optimized real-time performance using frame skipping and asynchronous processing for smooth CPU-based execution.

Vehicle Service Workshop Management Database | [GitHub](#) | *SQL, MySQL, Database Design*

2026

- Designed a relational database system for managing vehicle service operations, customer records, repair history, and maintenance workflows.
- Developed optimized SQL queries for service tracking, scheduling, reporting, and efficient data retrieval.
- Applied normalization techniques (1NF–3NF) and relational constraints to improve data integrity and scalability.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Java

Backend Development: FastAPI, REST APIs, WebSockets, SQLAlchemy, Pydantic

Databases: PostgreSQL, MySQL, Database Design, Relational Modeling

Software Engineering: Object-Oriented Programming, SOLID Principles, Dependency Injection, Design Patterns, Git, GitHub, Agile Development

Simulation & AI: Digital Twin Systems, Vehicle Telemetry, Computer Vision, Machine Learning, Fleet Analytics, Physics-based Simulation

Developer Tools: Docker, Linux, VS Code, Postman, Raspberry Pi

Libraries: OpenCV, NumPy, Pandas, Matplotlib

CERTIFICATIONS

Elements of AI

University of Helsinki

July 2025

Credential